Subsidizing Quality for the Poor: Equity and Competition in Health Insurance Markets

Kai Shen Lim

October 6, 2026

Motivation

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For **essential goods** (health care, education, housing, child care), society cares about **equality of access**, not just the **redistribution of income**

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HealthCare.gov	Federal Student Aid An OFFICE of the U.S. DEPARTMENT of EDUCATION	U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT	HARVARD Human Resources Childcare Scholarships
ACA tax credits	**Need-based College Pell Grants**	**Housing vouchers**	**Child care subsidies**

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Literature: *Level* (equal) subsidies ensures **affordability** and increases the poor's **participation**

Health insurance (Finkelstein, Hendren, Shepard 2019; Tebaldi 2025), college (Dynarski 2003; Dynarski et al. 2021), housing (Collinson and Ganong 2018; Bergman et al. 2024), child care (Baker, Gruber, Milligan 2008)

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- Subsidize a share of the **price gap** – e.g. if price ↑ \$10, social planner covers \$6

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MASSACHUSETTS HEALTH CONNECTOR	CSAC CALIFORNIA STUDENT AID COMMISSION	U.S. Department of Housing and Urban Development Small Area Fair Market Rents (SAFMRs)	MIT Human Resources Portable Child Care Subsidy Program
ConnectorCare extra subsidy for broad networks	**Cal Grant** larger grant at UC than at CSU	**Small Area FMRs** higher cap in pricier ZIP codes	**MIT Employee Benefits** MIT-affiliated childcare centers

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This paper: Understand **trade-offs** of **incremental subsidies** in **vertically differentiated** health insurance markets

This Paper

Setting: Massachusetts ConnectorCare (subsidized ACA marketplace)

- **Unique**: Separately use both *level* and *incremental* subsidies for the poor
- **Clean setting:** standardized insurance plans → compete in **prices and hospital network breadth**
- **Policy variation**: Removal of incremental subsidies in 2017

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Research Questions:

1. When is an incremental subsidy policy optimal? *(Theory)*
2. What is the incidence of the incremental subsidy? *(Reduced Form)*
3. What are the trade-offs of subsidizing the poor and alternate policies? *(Structural Model)*

Preview of Results

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Takeaway: Subsidize quality, not incremental prices

Related Literature and Contributions

1. **Subsidy design and competition in health insurance market**
   - **This paper**: equity in markets—how to subsidize higher-quality options and showing a **better alternate policy**

Decarolis 2015; Cabral, Geruso, Mahoney 2018; Jaffe and Shepard 2020; Tebaldi 2025; Finkelstein, Hendren, Shepard 2019; Shepard 2022, ...

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2. **In-kind transfers and welfarist vs. non-welfarist rationales**
   - **Existing lit:** cash beats in-kind transfers **unless** → market failures or paternalism as the reason to subsidize the poor
   - **This paper**: showing real-world policy cannot be rationalized by **welfarist planners** → implications for welfare analyses

Nichols and Zeckhauser 1982; Besley and Coate 1991; Currie and Gahvari 2008; Baicker, Chandra, Shepard 2023; Ambuehl et al. 2025

Outline

1. **Theory**
2. **Setting & Descriptive Evidence**
3. **Reduced Form**
4. **Structural Model, Welfare & Counterfactuals**

Theory

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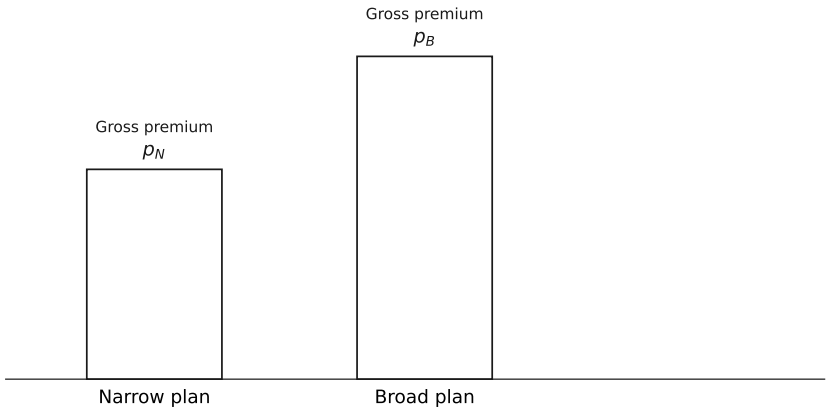
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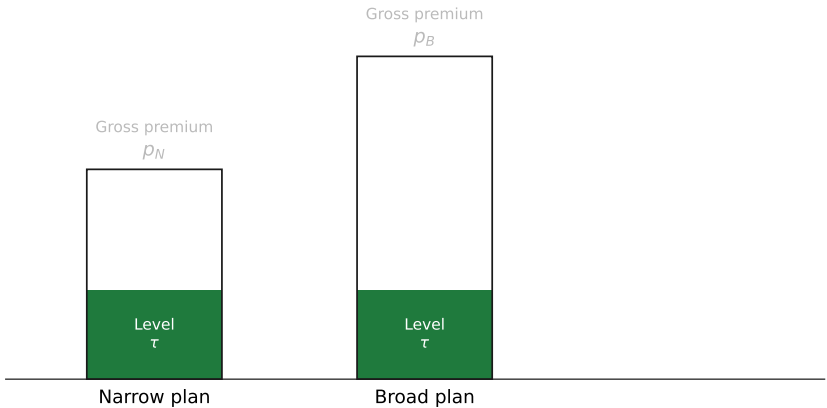
The planner subsidizes enrollees through two subsidy instruments:

- **Level** subsidy $\tau \geq 0$: the same amount towards **any** plan

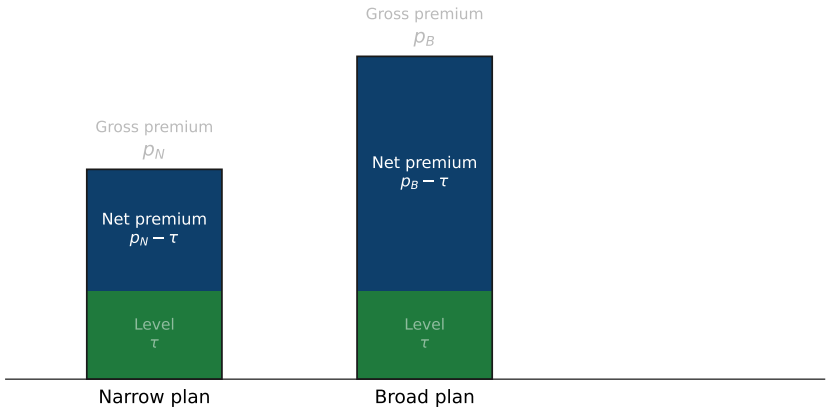
The Level Subsidy



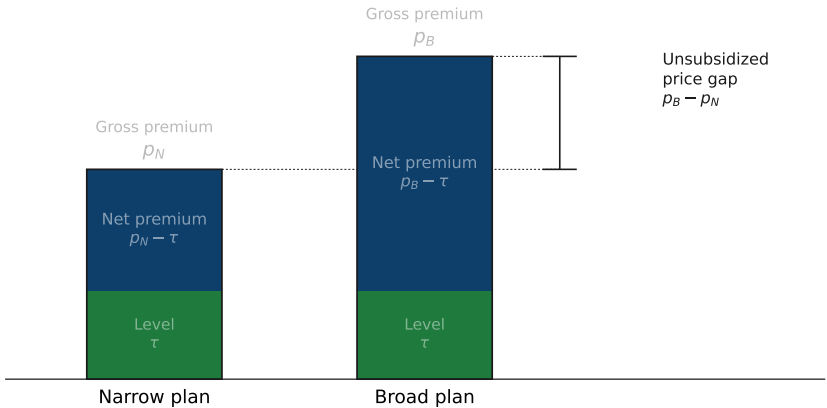
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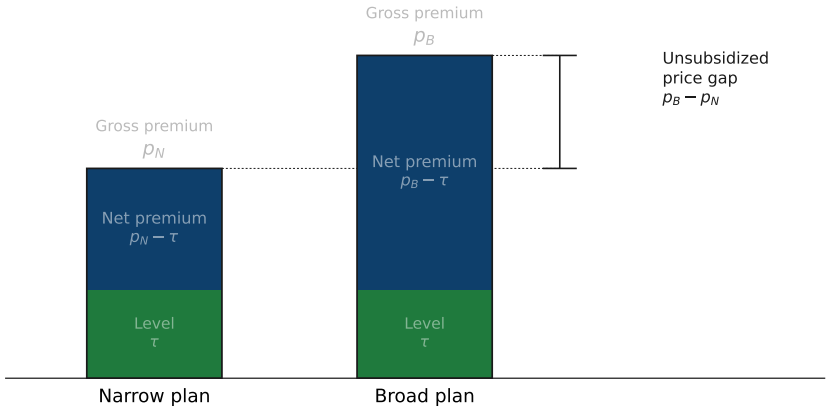
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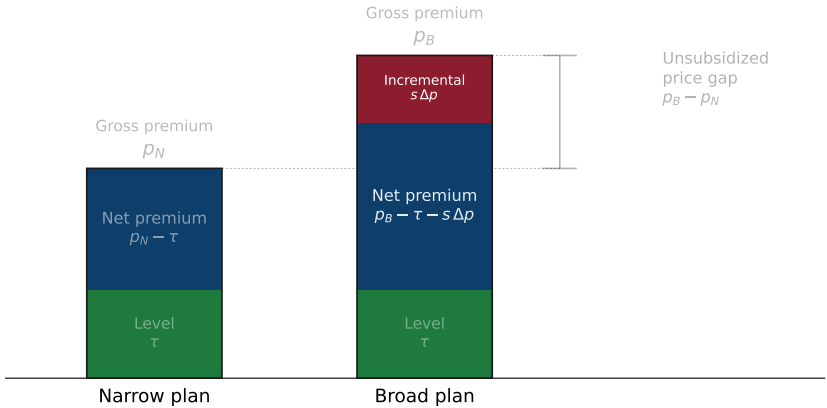
The planner subsidizes enrollees through two subsidy instruments:

- **Level** subsidy $\tau \geq 0$: the same amount towards **any** plan
- **Incremental** subsidy $s \in [0, 1]$: a share of the gross premium gap Δp

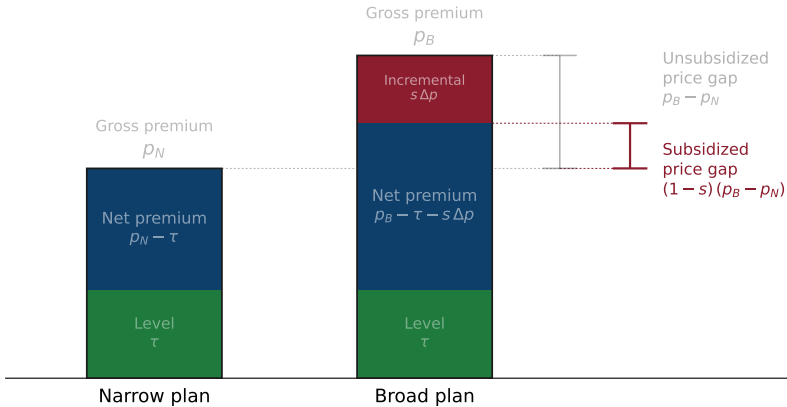
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Key Takeaway: Level subsidies akin to a lump-sum transfer; Incremental subsidies incentivize take-up of the high-quality plan

Theory Preview

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3. **Social planner.** Incremental subsidies preferred when the planner **values enrollment in the high-quality plan directly**
   - Otherwise level subsidies dominate: same transfer, no competition distortion

Step 1: Perfect Competition

Adverse selection: the sick value the high-quality plan more, so B's pool is sicker and costlier than average

Let enrollees $h \in \{S, H\}$. No outside option. **Insurers** price at average cost:

$$p_B \;=\; AC_B \;=\; c_{SB} \;-\; \underbrace{\delta_B}_{\text{cost gap}} \cdot \underbrace{\frac{D_{HB}(s)}{D_B(s)}}_{\substack{\text{healthy share} \\ \text{of } B\text{'s pool}}} \qquad \delta_B \equiv c_{SB} - c_{HB} \geq 0$$

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Adverse selection justifies $s > 0$ only if the subsidy makes B's pool healthier:

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$s > 0$ is justified if the subsidy raises healthy enrollment proportionally more than total enrollment

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Notice that a zero cost gap between sick and health $\delta_B = 0$: adverse selection cannot justify $s > 0$

Step 2: Imperfect Competition

Insurers now price at marginal cost plus a markup, $p_B = MC_B + 1/\eta_B$. How does s affect p_B?

$$\frac{dp_B}{ds} \propto \overbrace{-\delta_B \cdot \frac{\partial}{\partial s}\left(\frac{\partial D_{HB}/\partial s}{\partial D_B/\partial s}\right)}^{\text{adv. selection channel } (<0)} + \overbrace{\frac{\partial(1/\eta_B)}{\partial s}}^{\text{markup channel } (>0)}$$

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Adv. selection channel: as s increases, if the marginal enrollee is healthier $\rightarrow$ lower prices

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Markup channel: as s increases, enrollees bear only $1 - s$ of a price increase $\rightarrow$ prices go up

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p_B falls with s iff the cost channel outweighs the markup channel*

* Note: Under perfect risk adjustment ($\delta_B = 0$), only the markup channel remains

Social Planners

Define the planner's welfare objective over the subsidized **poor enrollees**:

$$W = -\underbrace{\lambda G}_{\text{fiscal cost}} + \underbrace{\omega CS}_{\text{consumer surplus}} + \underbrace{\mu D_B}_{\text{access}}$$

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Fiscal cost: government spending G on both subsidies, valued at $\lambda \geq 1$ per dollar, the cost of raising public funds

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Consumer surplus: **poor enrollees**' surplus valued at ω

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Access: enrollment in B itself

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Nests three planners:
efficiency $(\omega = 1,\ \mu = 0)$, inequality-averse $(\omega > 1,\ \mu = 0)$, Tobin $(\omega = 1,\ \mu > 0)$

Optimal Subsidy Trade-offs

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$$\frac{\partial W}{\partial \tau} = (\omega - \lambda) \cdot \underbrace{\left(D_B + D_N\right)}_{\text{all enrollees}}$$

Level subsidy: a dollar to every enrollee, valued at ω, at a fiscal cost of λ, with no effect on plan choice: a lump-sum transfer from the government to consumers

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Incremental subsidy: three terms, at fixed prices

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Transfer: the same $\omega - \lambda$ per dollar, but only on the Δp that enrollees already in B receive

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Access: the planner's own valuation of each switcher the subsidy draws into B

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Subsidy on switchers: the state pays $s \cdot \Delta p$ for each switcher

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Level subsidy weighs CS vs. cost for **all enrollees**
Incremental subsidy cares about **enrollment in B**

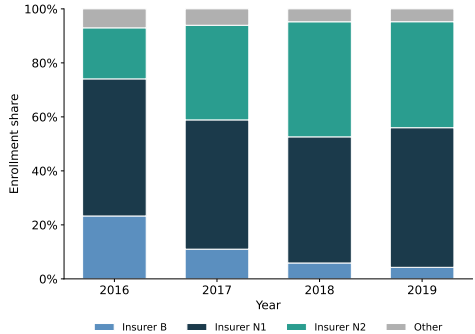
Setting and Descriptive Evidence

Setting

ConnectorCare: subsidized segment of the Massachusetts ACA marketplace

Key facts:

- Three insurers hold most of the market: Insurer B (broad - covers 'star' hospitals) vs. Insurers N1/N2 (narrower)
- Plans are **standardized HMOs** with no deductible: compete on **prices** and **breadth of hospital network**
- **Five income groups**: below 100 percent to 300 percent FPL



Enrollment share by insurer, 2016–2019

Incremental subsidy motivation



Policy & Strategy: Recent Milestones

	Description
Products and programs	[illegible]
	[illegible]
	[illegible]
	▪ Starting in the 2022 Seal of Approval, adjusted the application of state premium subsidies in ConnectorCare to ensure that the lowest income enrollees (disproportionately immigrants) have meaningful carrier choice
Promoting enrollment	[illegible]
	[illegible]
	[illegible]

From Gross Premium to Revenue and Net Premium

Insurer sets gross premium

$$p_j \times \psi_{j\,r(i)}$$

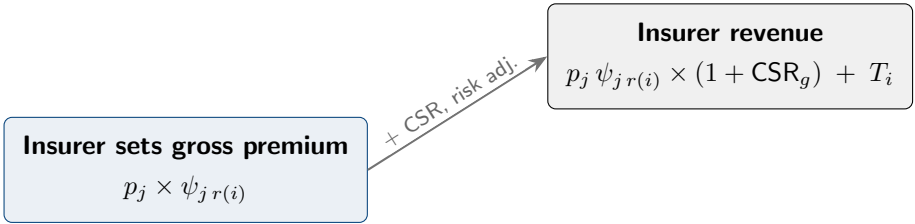
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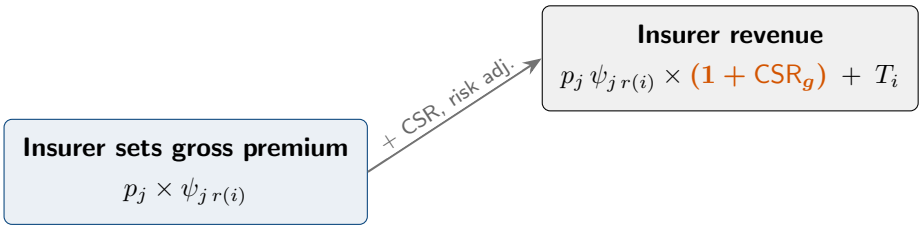
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Rating factors $\psi_{j\,r(i)}$: ACA age curve (oldest pay at most 3 times the youngest) $\times$ regional factor

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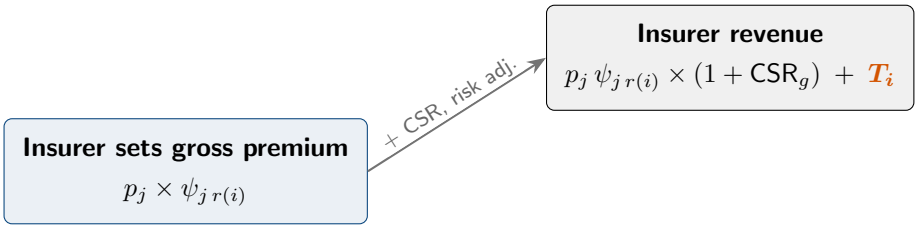


From Gross Premium to Revenue and Net Premium



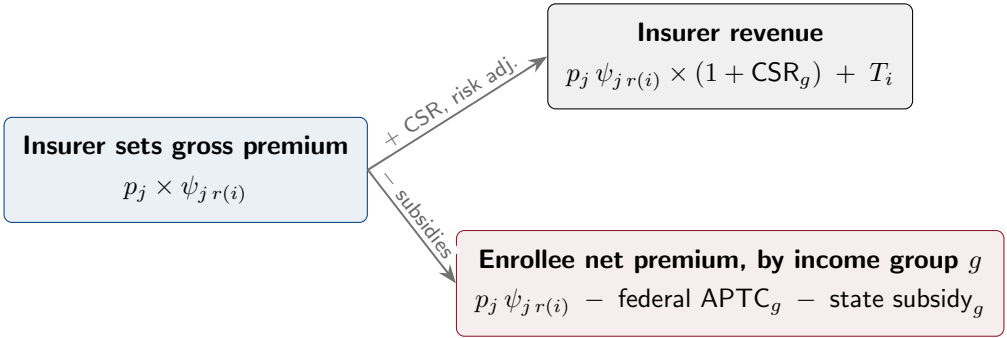
Cost-sharing reductions (CSR): ConnectorCare plans carry lower copays than a standard silver plan. The state and federal government reimburse the insurer for the richer benefit design as a fixed share of the premium, set by income group (31 to 34 percent in 2016)

From Gross Premium to Revenue and Net Premium

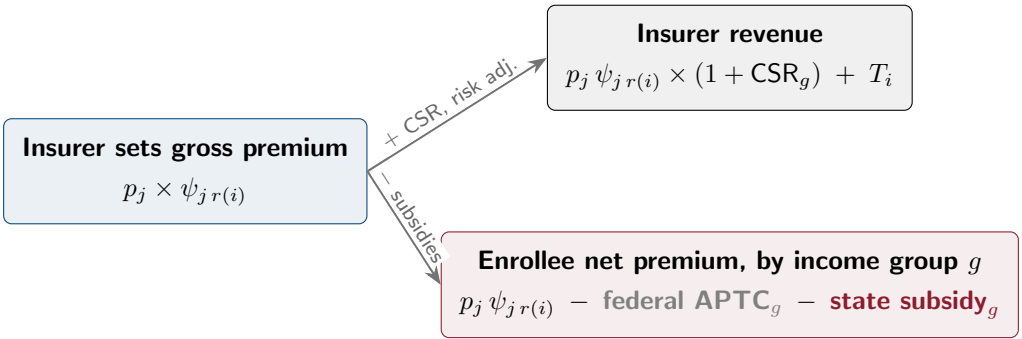


Risk adjustment T_i: transfer across insurers that pays for enrollee i's risk score above the market average, at the average premium: $T_i = (h_i/\bar{h} - 1)\,\bar{P}$. Compensates insurers for sicker enrollees; zero-sum across insurers

From Gross Premium to Revenue and Net Premium



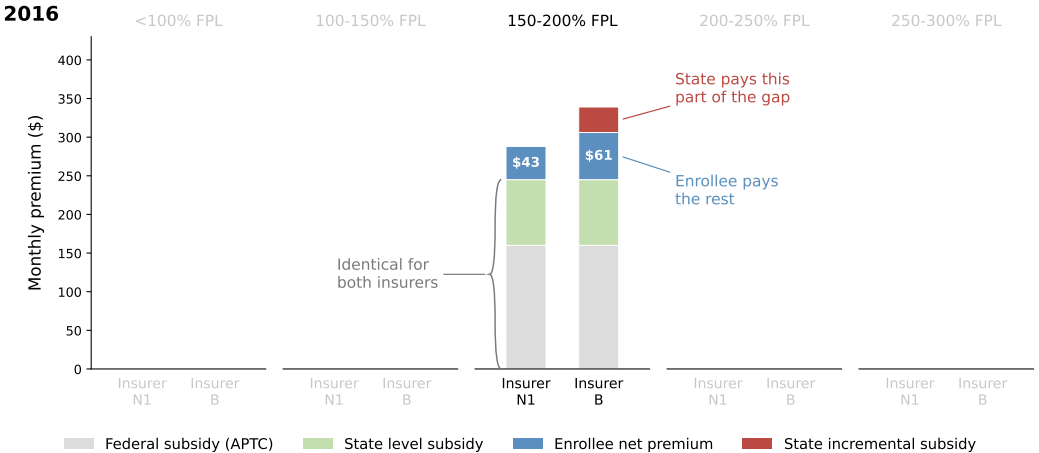
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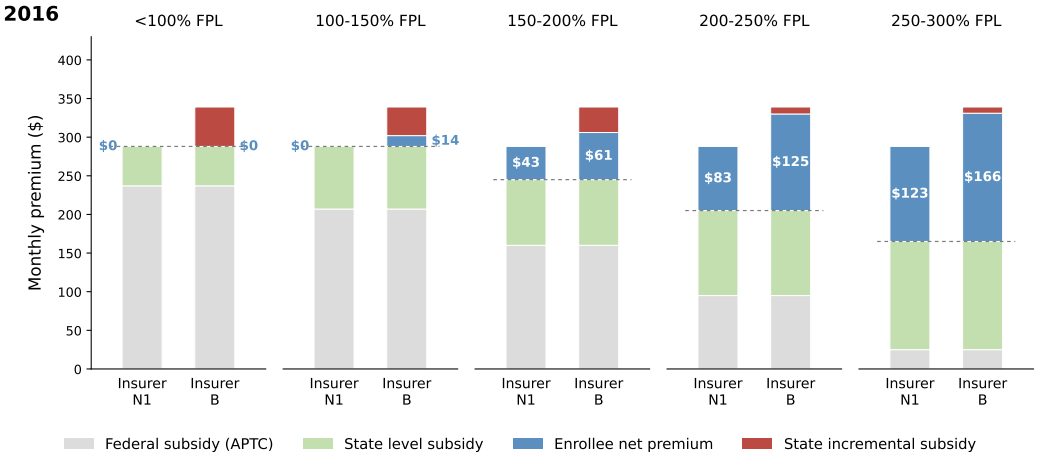
Federal APTC$_g$: benchmark silver premium minus a share of income that rises with g; the same dollars toward any plan

State subsidy$_g$: level (sets what group g pays for the cheapest plan) + incremental (share s_g of the gap to Insurer B, falling with income in 2016). Net premiums do not vary by age

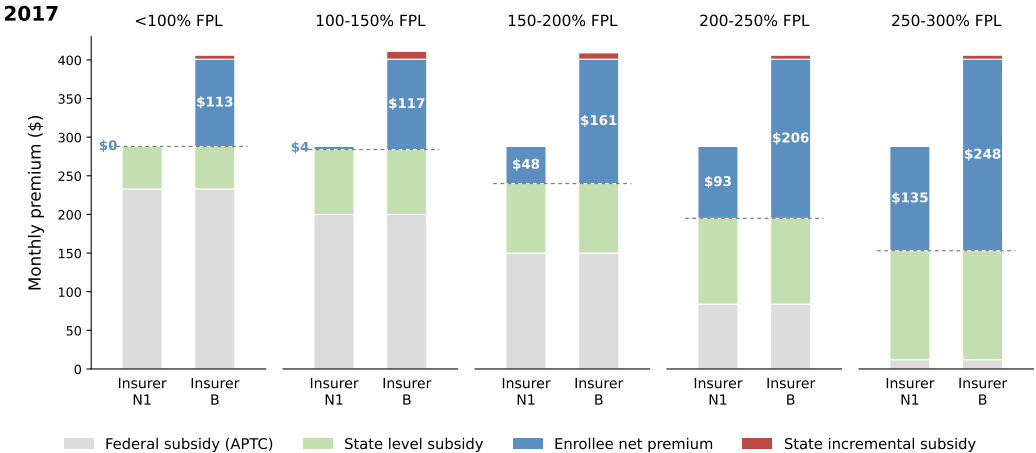
Premium Decomposition



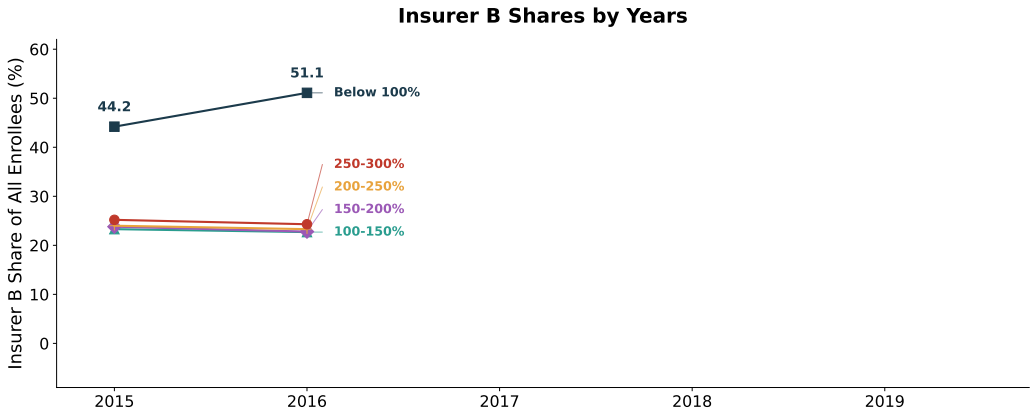
In 2016, ConnectorCare implemented progressive incremental subsidies



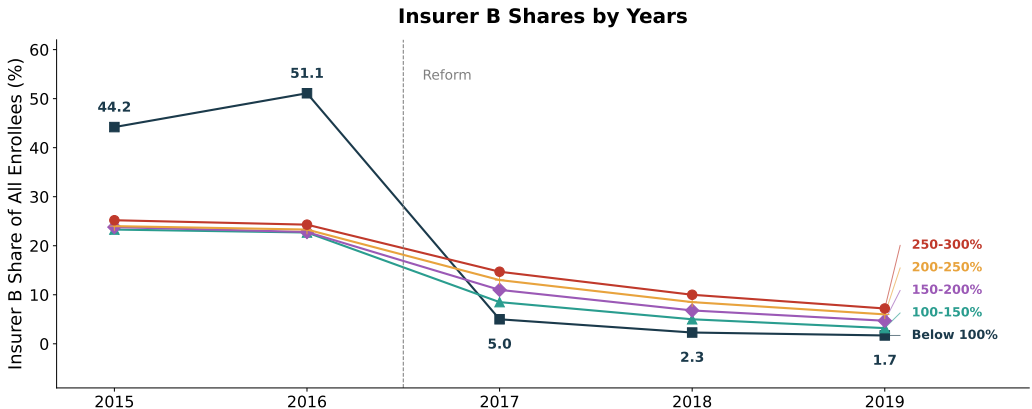
In 2017, ConnectorCare mostly removed incremental subsidies



In 2016, Insurer B's market share (of a specific income group): 22 to 51 percent



In 2017, Insurer B's market share fell significantly



Reduced Form

Enrollment: 2017 Subsidy Removal

Question: How did removing incremental subsidies affect enrollment in Insurer B?

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Design: DiD across income groups, 250–300% FPL as control, new and continuing enrollees separately

$$\text{Insurer B}_{igrt} = \sum_{g} \beta_g \big(\mathbf{1}\{G_i = g\} \times \text{Post-2017}_t \big) + \gamma_{gr} + \tau_t + \varepsilon_{igrt}$$

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β_g: post-2017 change in Insurer B enrollment for income group g, relative to 250–300% FPL

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Fixed effects: income group $\times$ region absorbs local taste for Insurer B; year absorbs statewide shocks

Enrollment: 2017 Subsidy Removal

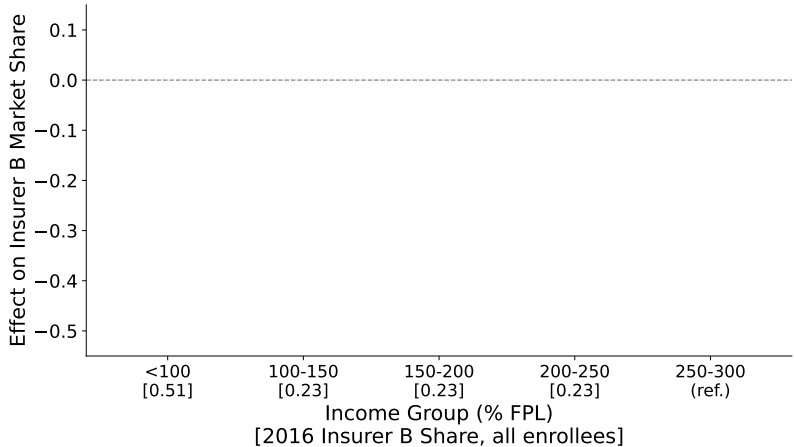
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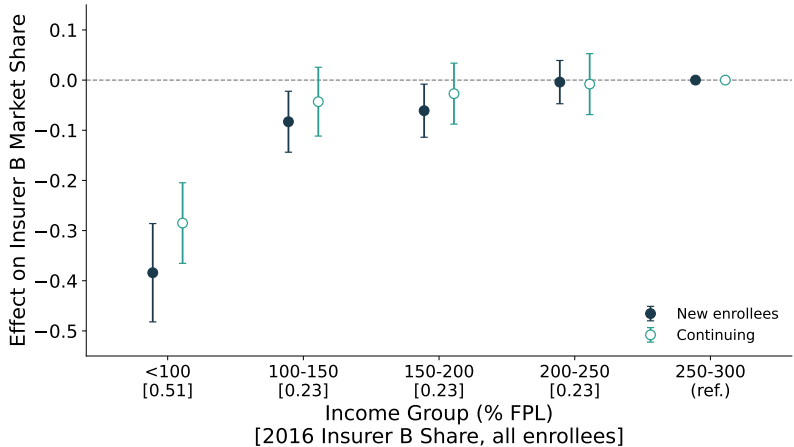
$$\text{Insurer B}_{igrt} = \sum_{g} \beta_g \big(\mathbf{1}\{G_i = g\} \times \text{Post-2017}_t \big) + \gamma_{gr} + \tau_t + \varepsilon_{igrt}$$

Identifying assumption: absent the removal, income groups within a region would have trended similarly

Coefficient plot of β_g



The poor left Insurer B



Selection: Who Left Insurer B?

Question: Were the enrollees who left Insurer B healthier than those who stayed?

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Outcome: enrollee's concurrent HCC risk score, from same-year claims

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δ_g: post-2017 change in Insurer B enrollees' risk relative to Insurer N1/N2 enrollees', within income group g. Control group: same-income enrollees in the narrow insurers

Selection: Who Left Insurer B?

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$\ldots$: Insurer B $\times$ income group and income group $\times$ post-2017 terms, absorbing level differences and group-level health trends

Selection: Who Left Insurer B?

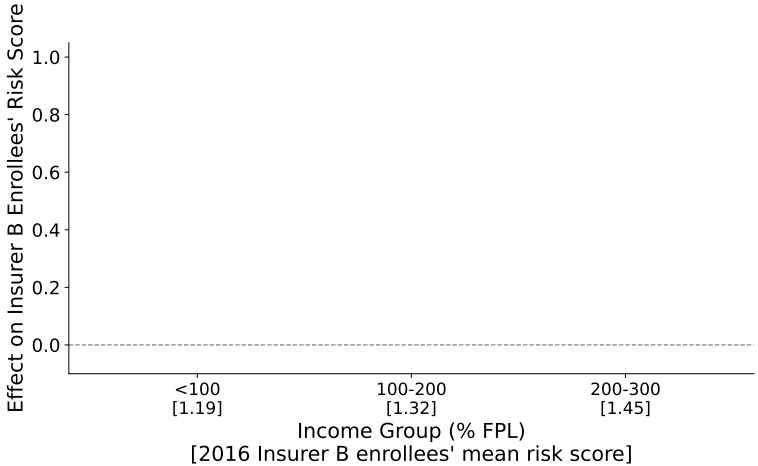
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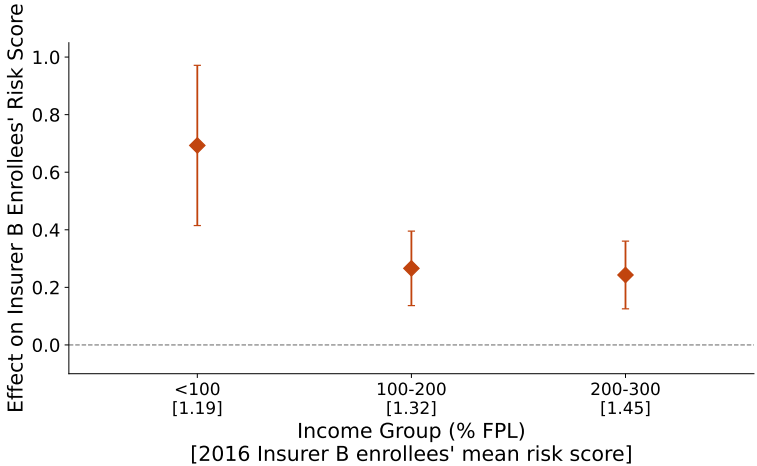
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Identifying assumption: absent the removal, Insurer B's and Insurer N1/N2's enrollees within an income group would have trended similarly in risk

Coefplot of δ_g



Insurer B left with unhealthier enrollees



Reduced Form: Summary

Removing the **progressive incremental subsidies** in 2017 led to:

1. **Enrollment** (Price sensitivity): The poor left Insurer B

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Justification for addressing **adverse selection** exists alongside **access motives**

Structural Model

Structural Model, Welfare & Counterfactuals

Demand: Micro-BLP on enrollment data, APCD claims as micro data Demand

- Price sensitivity by income group; health and distance sorting into Insurer B

Insurer Cost: $c_{ij} = \hat{c}_i \cdot \delta_j$ Costs

- Predicted cost $\hat{c}_i$ from claims; insurer cost effects δ_j from plan switchers

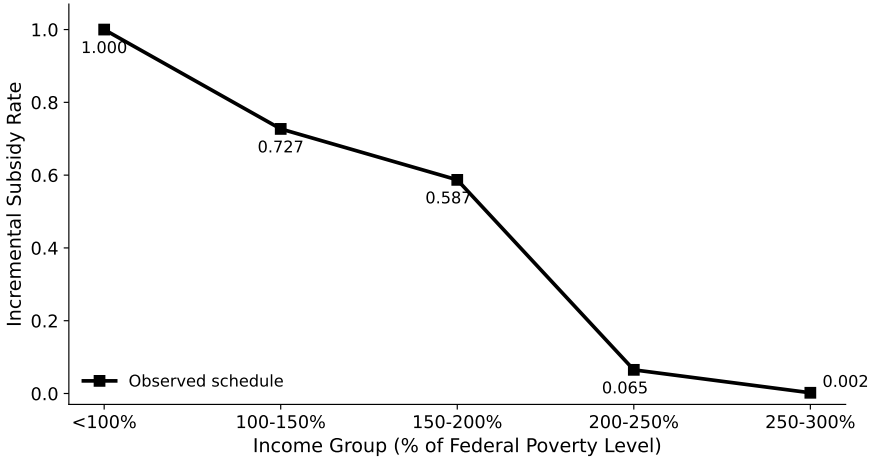
Insurer Pricing: set gross premium Pricing

- Revenue $= p_j \cdot \psi_{jr} \cdot (1 + \text{CSR}_g) + T_i$: gross premium $\times$ fixed age and regional rating factors, plus cost-sharing reduction payments and risk adjustment

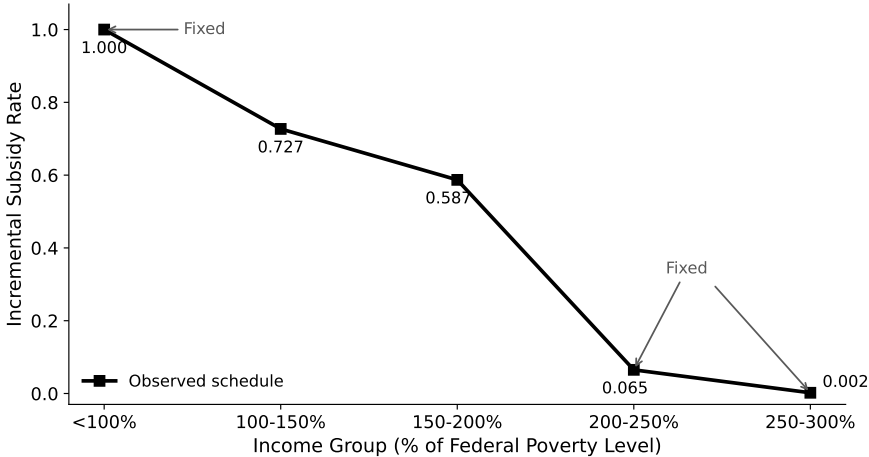
Equilibrium: Nash-Bertrand in gross premium Equilibrium Model Fit

- Maps subsidy policies into premiums, enrollment, profits, government cost

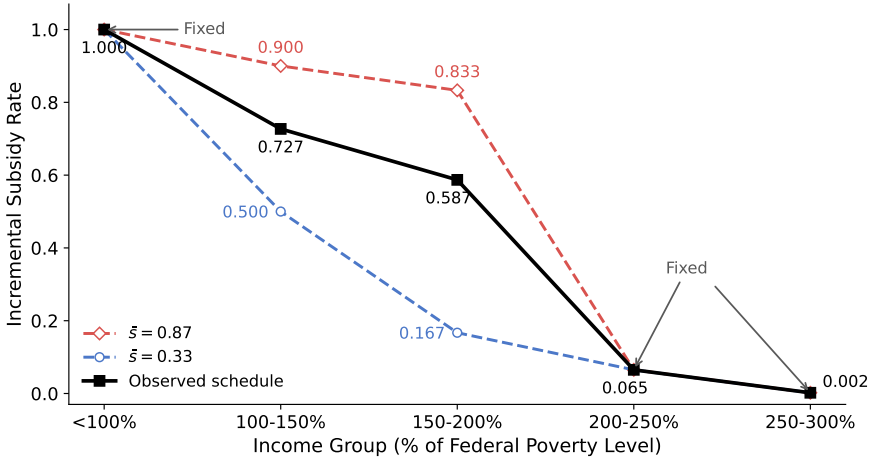
Varying incremental subsidy rate s



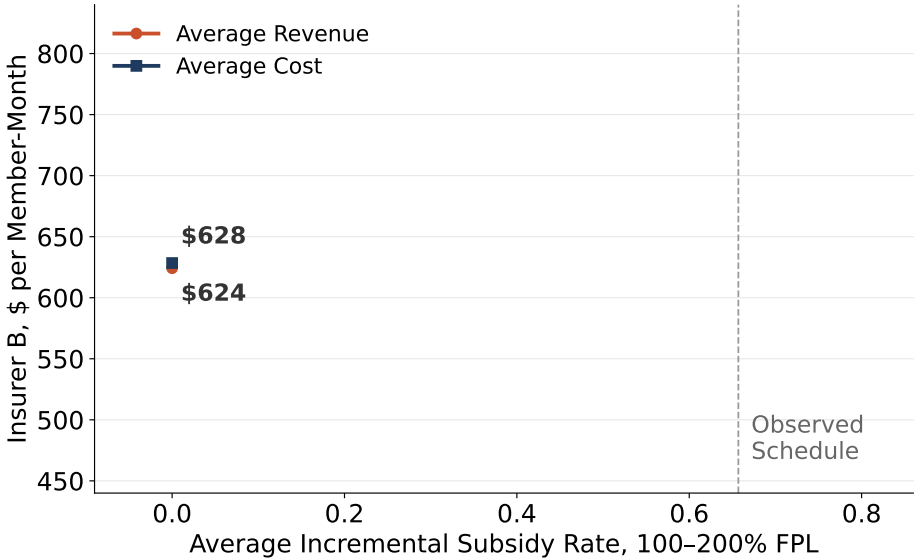
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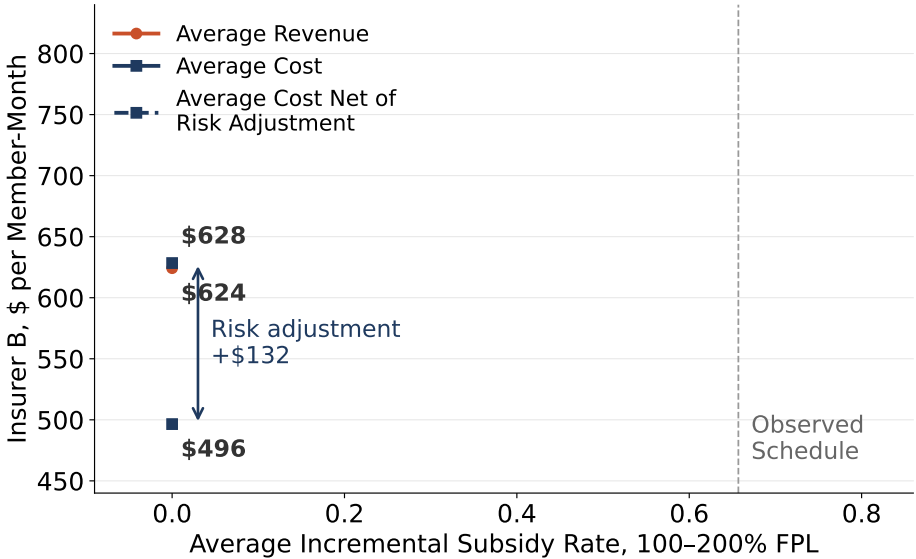
Varying incremental subsidy rate s



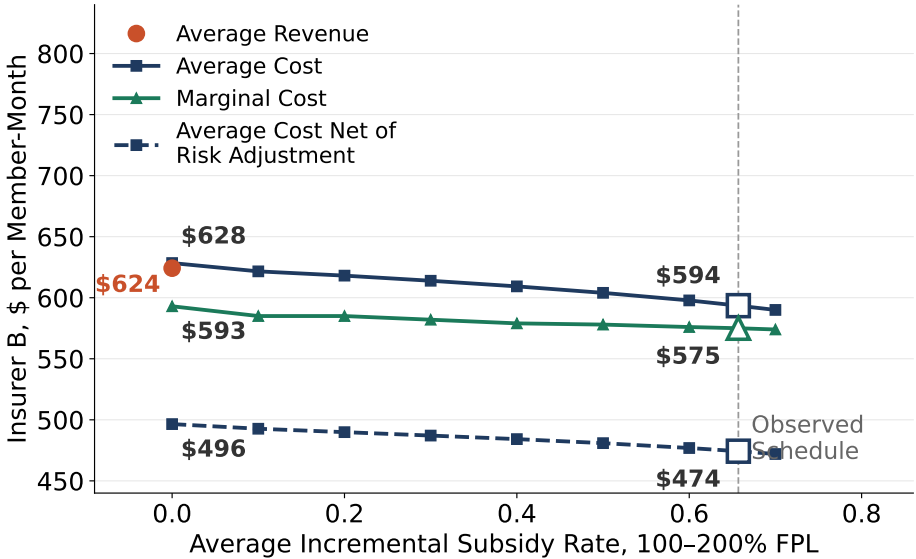
How do incremental subsidies affect Insurer B?



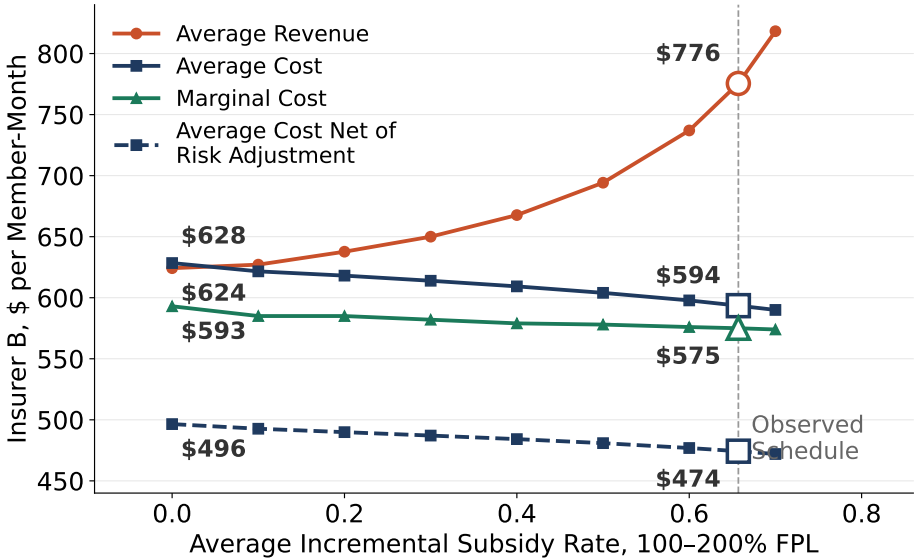
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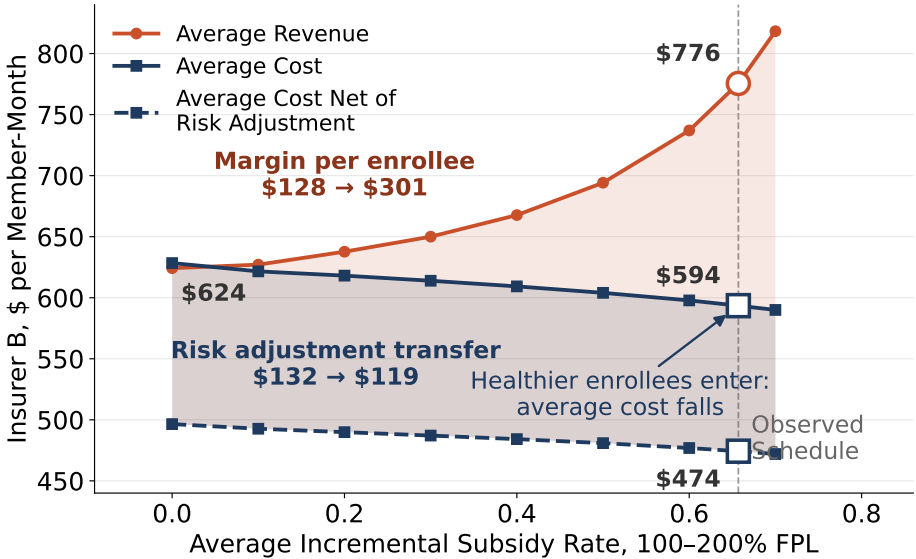
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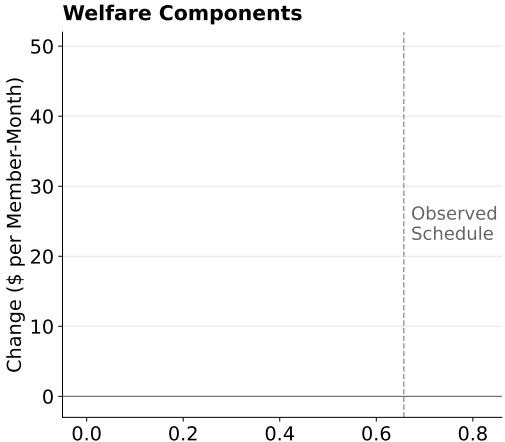
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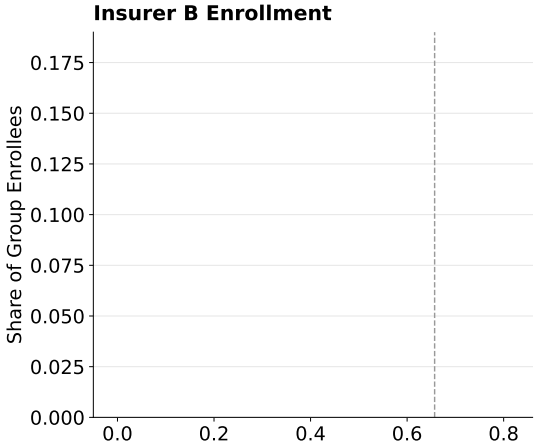


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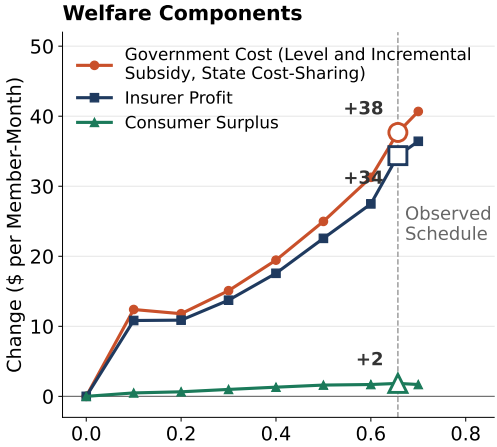
Welfare and Enrollment

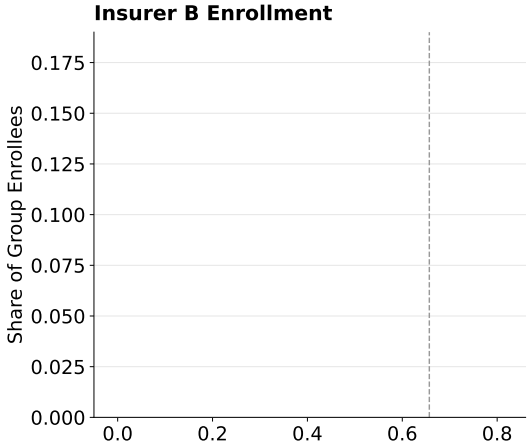




Average Incremental Subsidy Rate, 100–200% FPL

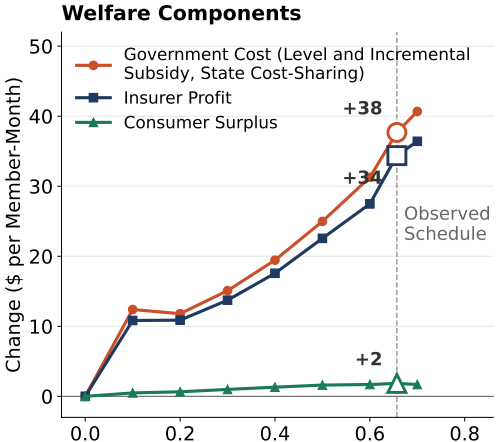
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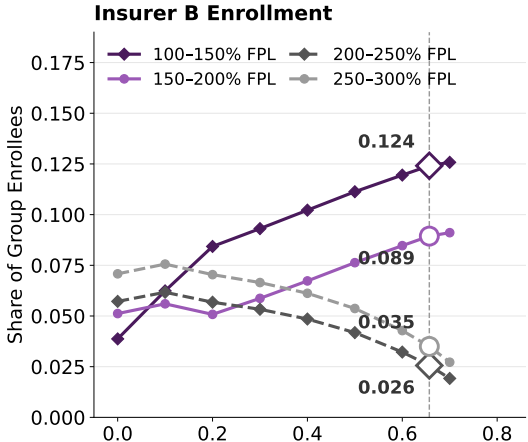




Average Incremental Subsidy Rate, 100–200% FPL

Welfare and Enrollment





Average Incremental Subsidy Rate, 100–200% FPL

Welfare Simulations: Summary

1. **Adverse selection justifies part of the incremental subsidy but markups soar**
   - Healthier enrollees enter Insurer B: its average cost falls \$628 $\rightarrow$ \$594 per member-month
   - But its revenue rises \$624 $\rightarrow$ \$776; of each incremental dollar, 91 cents is insurer profit

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Welfare Simulations: Summary

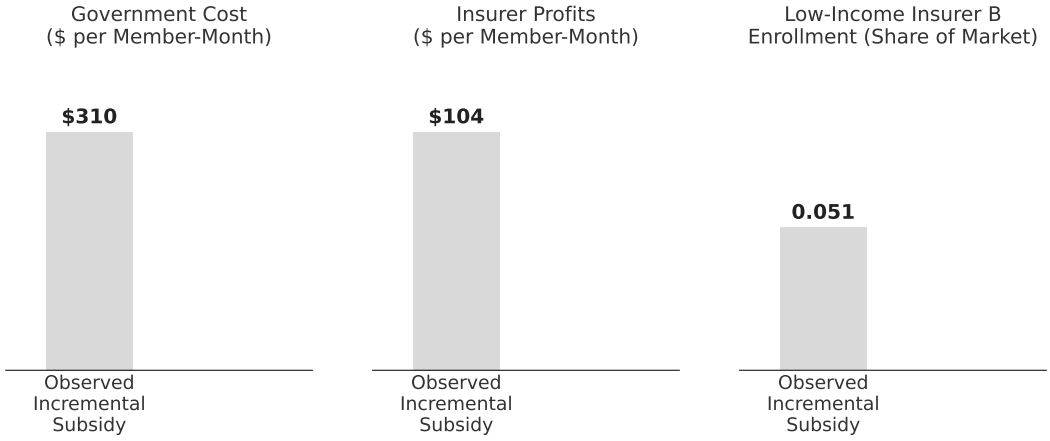
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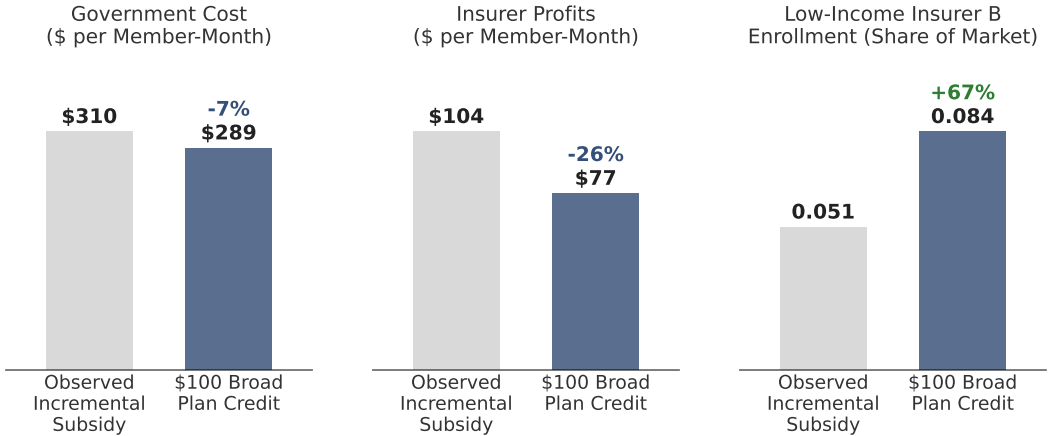
4. **Is there a policy that yields Insurer B enrollment without paying for markups?**

Counterfactual Policy: A $100 Broad Plan Credit



Remove the incremental subsidy above poverty; pay $100 per member-month toward Insurer B's premium for 100–200% FPL enrollees

Counterfactual Policy: A $100 Broad Plan Credit



Two-thirds more low-income enrollment in Insurer B; insurers lose profits

Conclusion

Conclusion

Subsidize quality, not incremental prices

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Subsidize quality, not incremental prices

HEALTH INSURANCE	COLLEGE	HOUSING	CHILD CARE
MASSACHUSETTS HEALTH CONNECTOR	CSAC CALIFORNIA STUDENT AID COMMISSION	U.S. Department of Housing and Urban Development Small Area Fair Market Rents (SAFMRs)	MIT Human Resources Portable Child Care Subsidy Program
ConnectorCare	**Cal Grant**	**Small Area FMRs**	**MIT Employee Benefits**
share of the premium gap to the broad plan	share of the tuition gap between UC and CSU	share of the rent gap to pricier ZIP codes	need-based share of tuition at MIT's own centers

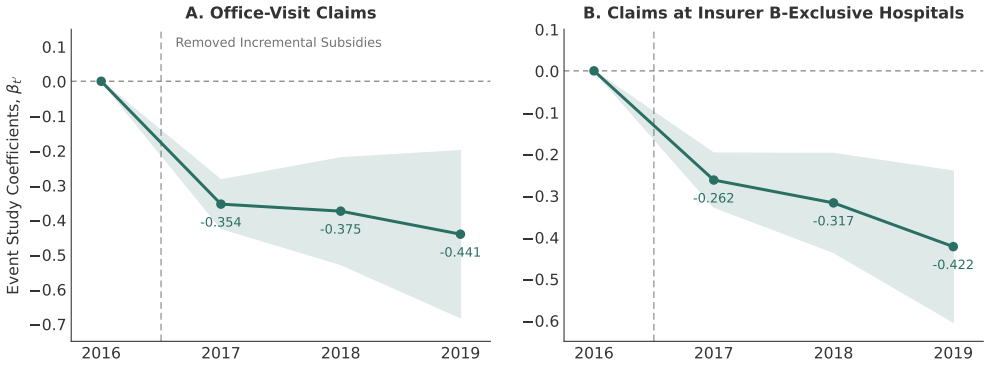
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~~share of the premium gap to the broad plan~~ fixed credit for choosing the broad plan	~~share of the tuition gap between UC and CSU~~ fixed award for enrolling at UC	~~share of the rent gap to pricier ZIP codes~~ fixed bonus for leasing in a high-opportunity ZIP	~~need-based share of tuition at MIT's own centers~~ fixed stipend for enrolling at an MIT center

Appendix

Appendix: Care Disruption Among the Poor Who Left Insurer B



Event study, below-poverty switchers vs. below-poverty enrollees already in Insurer N1 or N2 in 2016; person and year FE; claims per member-month; 95% CI, SEs clustered by region × year.

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Appendix: Demand Specification

Enrollee i in income group g, region r, year t chooses insurer j:

$$u_{ijgrt} = \underbrace{\alpha_g \tilde{p}_{jgrt}}_{\text{net premium, by income}} + \underbrace{\beta \mathbf{1}\{P_{jrt} = \min_k P_{krt}\}}_{\text{lowest-cost insurer in region}} + \underbrace{\xi_{jr} + \xi_{jg}}_{\text{insurer quality}} + \underbrace{\theta_h h_i \mathbf{1}\{j = B\}}_{\text{health sorting}} + \underbrace{\theta_d d_i \mathbf{1}\{j = B\}}_{\text{distance to } B\text{-exclusive hospital}}$$

$$+ \underbrace{\zeta_{i,n(j)} + (1-\rho)\varepsilon_{ijgrt}}_{\text{nested logit: N1 and N2 in one nest}}$$

- Sample: new enrollees, 2016–2019; market = region × year × income group (160 markets)
- Micro moments: mean risk score and mean distance among Insurer B enrollees
- Instrument: statewide gross premiums × income group × year

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Appendix: Demand Identification

Parameter	**Source of variation**
α_g price sensitivity	2017 subsidy removal raised Insurer B's net premium differently by income group
ρ nest correlation	N1 vs. N2 log odds response to their net premium gap across years with different pass-through (full in 2018; 27–41% in 2016)
β lowest-cost premium	Regions where the cheapest insurer changed across years while net premiums moved little
$\theta_h,\ \theta_d$	Risk-score and distance micro moments from claims

Untargeted check: share of Insurer B leavers going to N1 vs. N2

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Appendix: Demand Estimates

Price sensitivity α_g (per \$100/month)	
Below 100% FPL	-2.72 (0.10)
100–150% FPL	-1.42 (0.06)
150–200% FPL	-1.23 (0.05)
200–250% FPL	-1.11 (0.05)
250–300% FPL	-0.93 (0.05)
Lowest-cost premium β	0.054 (0.017)
Health sorting θ_h	0.200 (0.040)
Distance θ_d	-0.265 (0.125)
Nest correlation ρ	0.80 (pinned)

Own-price semi-elasticity (% per \$1 net premium)	
Insurer N1	-1.45
Insurer B	-0.93
Insurer N2	-4.16

- Poor are 3× as price sensitive as 250–300% FPL
- Insurer B least elastic: alone in its nest

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Appendix: Insurer Cost

Cost to insurer j of covering enrollee i:

$$c_{ij} = \underbrace{\hat{c}_i}_{\substack{\text{predicted cost} \\ \text{(age, sex, income, risk score)}}} \times \underbrace{\delta_j}_{\substack{\text{insurer} \\ \text{cost effect}}}$$

δ_j from within-person cost changes of enrollees who switch insurers:

$$\mathbb{E}[c_{it} \mid \cdot] = \exp\left(\lambda_B \,\text{Insurer B}_{it} + \lambda_{N2} \,\text{Insurer N2}_{it} + \varrho\, h_{it} + \upsilon_i + \chi_t\right), \qquad \delta_j = e^{\lambda_j},\ \delta_{N1} = 1$$

- Poisson PML; person and year FE; Insurer N1 base

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Appendix: Cost Estimates

	Insurer N1	Insurer B	Insurer N2
Observed, 2016			
Average cost ($/month)	277.8	552.5	299.2
Mean risk score	0.96	1.36	0.94
Person-years	62,901	28,432	24,362
Insurer cost effect δ_j, 2016–2019			
Cost effect	base	1.50	1.33
		(0.046)	(0.121)

- Same person costs 50% more at Insurer B, 33% more at Insurer N2
- Insurer B's raw cost gap ($2\times$) is half network, half sicker enrollees

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Appendix: Insurer Revenue and Profit

Insurer j sets one statewide gross premium p_j; revenue per enrollee is not p_j:

$$\pi_j = \sum_i \mathcal{S}_{ij} \Big[\underbrace{p_j \, \psi_{jr(i)}}_{\substack{\text{age- and region-} \\ \text{rated premium}}} \big(1 + \underbrace{\text{CSR}_{g(i)}}_{\substack{\text{cost-sharing} \\ \text{loading}}}\big) - \underbrace{c_{ij}}_{\text{cost}} + \underbrace{T_i}_{\substack{\text{risk} \\ \text{adjustment}}} \Big]$$

- **Rating factors** ψ_{jr}: ACA age curve (3:1 cap) and regional factor, set by regulation
- **CSR payments**: state reimburses the richer benefit design; proportional to premium within income group, so it enters the pricing margin
- **Risk adjustment** $T_i = (h_i/\bar{h} - 1)\,\bar{P}$, with $\bar{P}$ held at its observed level (as in Tebaldi, 2025)

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Appendix: Equilibrium

- Insurers set gross premiums simultaneously; Nash equilibrium in (p_{N1}, p_B, p_{N2}); entry fixed
- Lowest bid in each region is the subsidy benchmark: its enrollees pay the income contribution alone; recomputed at every point of the solution
- Each policy (τ, s) re-solves all three gross premiums: outputs are premiums, enrollment, profits, government cost

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Appendix: Model Fit, 2016

	Data	Model
Average cost ($/member-month)		
Insurer N1	277.8	275.4
Insurer B	552.5	562.9
Insurer N2	299.2	356.8
Gross premium ($/month)		
Insurer N1	240	273
Insurer B	274	484
Insurer N2	247	273
Market shares		
Insurer N1	0.577	0.584
Insurer B	0.221	0.154
Insurer N2	0.202	0.261

- Costs fit; N1 and N2 gross premiums within 14% (no rate review in model)
- Insurer B's filed premium is below its static best response: counterfactuals re-solve all premiums and hold conduct fixed

Appendix: Planners in the Simulation

Question: which planner would choose the observed incremental subsidy?

Nested welfare function: $W(s) = \sum_g \omega_g \, CS_g(s) \; - \; \lambda \, G(s) \; + \; \mu \, D_B(s)$

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Efficiency: $\omega_g = 1, \; \mu = 0$

- Every dollar counts the same; the only wedge is the cost of public funds, $\lambda = 1.3$

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- Every dollar counts the same; the only wedge is the cost of public funds, $\lambda = 1.3$

Utilitarian: $\omega_g = (y_{\text{med}}/y_g)^\gamma, \; \mu = 0$

- CRRA marginal utility of income, normalized at median income; γ is inequality aversion. OMB uses $\gamma = 1.4$ for U.S. benefit-cost analysis (Office of Management and Budget, 2023)

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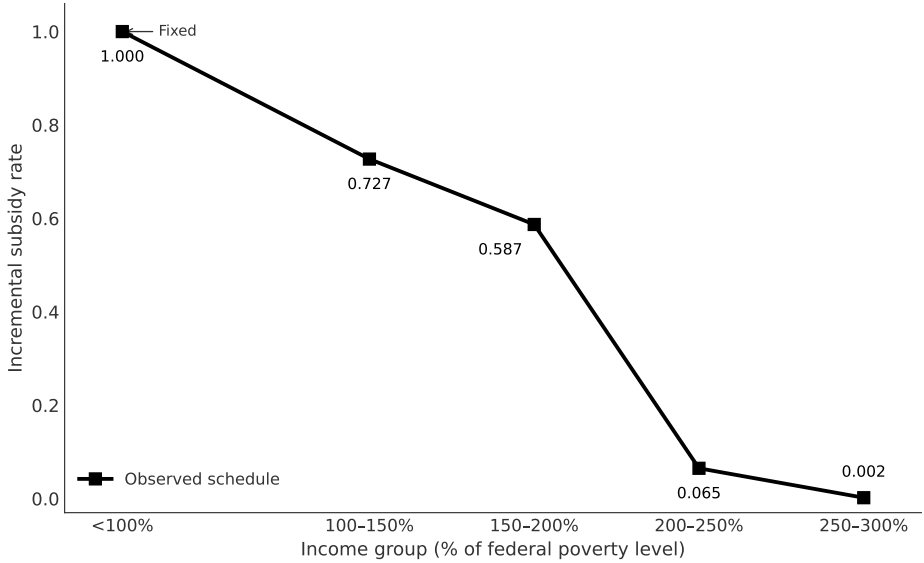
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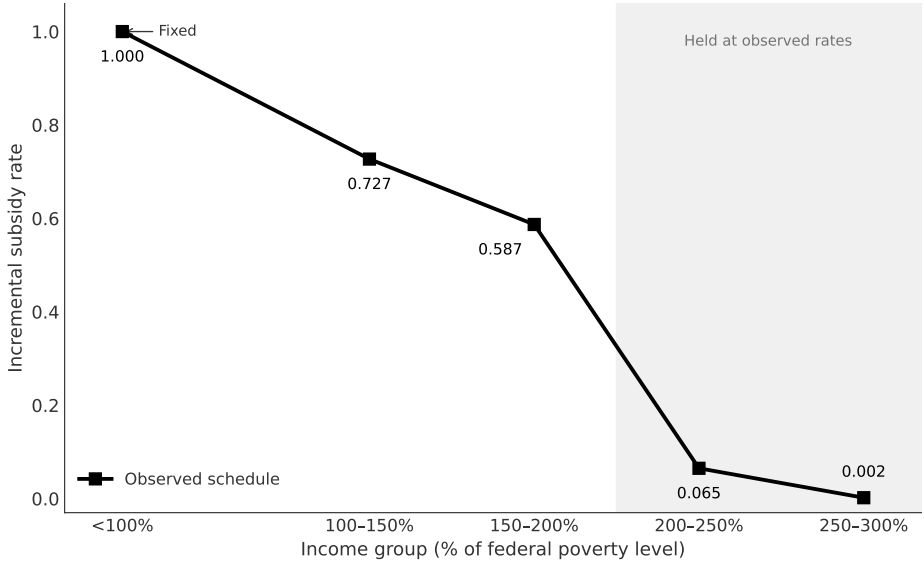
Tobin: $\omega_g = 1, \; \mu > 0$

- Values 100–200% FPL enrollment in Insurer B, D_B, beyond standard welfarist measures

Appendix: Recovering μ from the Observed Schedule

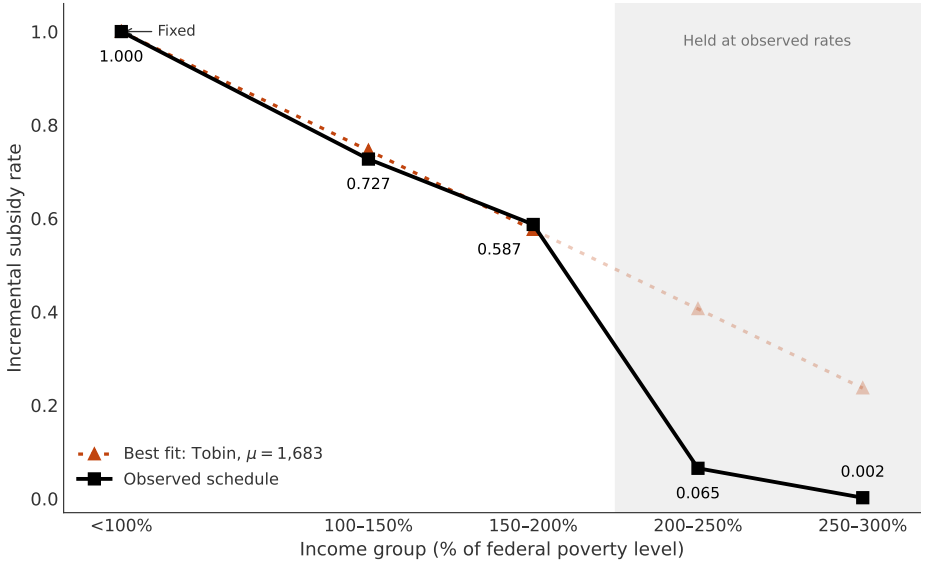


Appendix: Recovering μ from the Observed Schedule



Above 200% FPL the rate is left the primary essentially unchanged, holding the current

Appendix: Recovering μ from the Observed Schedule



Appendix: Which Planner Chooses the Incremental Subsidy?

	Simulated welfare ($/member-month)			
Planner	Observed policy	Level only	Increased level	Prefers

Three policies, each a simulated equilibrium with every gross premium re-solved. **Observed policy**: the level and incremental subsidy. **Level only**: the incremental subsidy removed. **Increased level**: level only plus a lump sum to 100–200% FPL enrollees equal to the incremental subsidy's cost, $34.7 per member-month

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Appendix: Which Planner Chooses the Incremental Subsidy?

	Simulated welfare ($/member-month)			
Planner	Observed policy	Level only	Increased level	Prefers
Efficiency	64.4	**107.8**	97.4	Level only

Efficiency planner prefers level only: the incremental subsidy costs $34.7 and returns $1.7 of consumer surplus

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Appendix: Which Planner Chooses the Incremental Subsidy?

	Simulated welfare ($/member-month)			
Planner	Observed policy	Level only	Increased level	Prefers
Efficiency	64.4	**107.8**	97.4	Level only
Utilitarian ($\gamma = 1.4$)	40.8	**82.9**	79.7	Level only
Utilitarian ($\gamma = 5$)	270.0	306.8	**348.8**	Increased level

Utilitarian planner never prefers the observed policy: at the OMB value it prefers level only; at extreme aversion, increased level subsidies, which leave every income group better off at the same cost

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Utilitarian ($\gamma = 1.4$)	40.8	**82.9**	79.7	Level only
Utilitarian ($\gamma = 5$)	270.0	306.8	**348.8**	Increased level
Tobin ($\mu = 3{,}821$)	**255.6**	197.8	187.4	**Observed**

Only the Tobin planner prefers the observed policy: at $\mu = \$3{,}821$, 6.6 times Insurer B's premium and about 470 times enrollees' own valuation of access, the increment beats both alternatives

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Appendix: Which Planner Chooses the Incremental Subsidy?

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Utilitarian ($\gamma = 5$)	270.0	306.8	**348.8**	Increased level
Tobin ($\mu = 3{,}821$)	**255.6**	197.8	187.4	**Observed**

Adverse selection lowers the valuation the policy needs: with risk adjustment made perfect, the recovered μ rises from $3,821 to $8,610. Selection does part of the planner's work; the rest is access

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